

5. Constructor overloading CO1

An organization is maintaining the data of employee according to cadre of employee with following parameters: name, id, designation, salary, promotion status. Apply the constructor overloading to implement it.

1. Steps of the Program

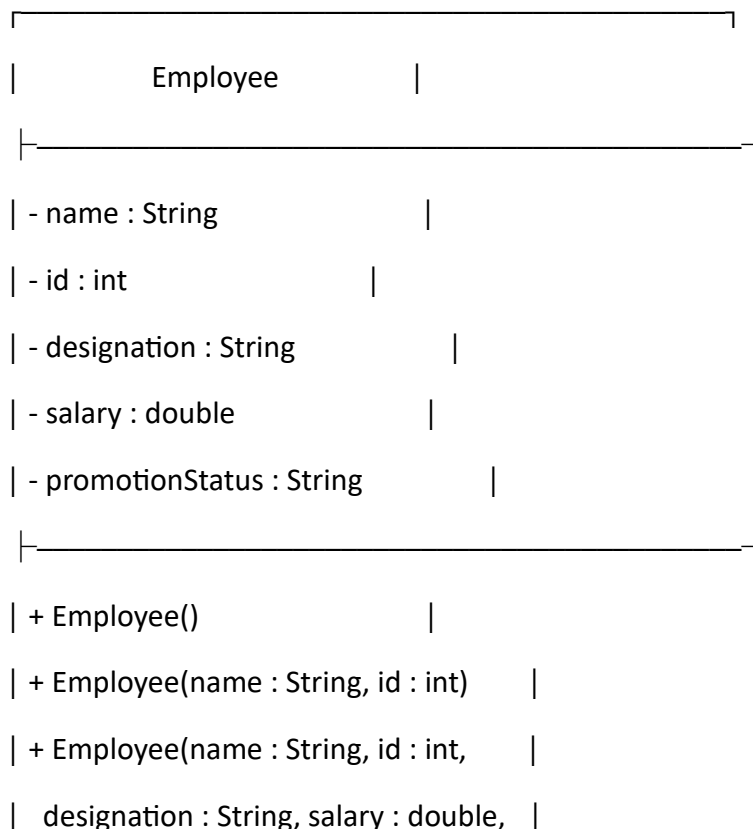
1. Create a class named Employee.
 2. Declare employee details:
 - name
 - id
 - designation
 - salary
 - promotionStatus
 3. Create a **default constructor** with no parameters.
 4. Initialize default values in the default constructor.
 5. Create a **2-parameter constructor** with name and id.
 6. Assign default values for designation, salary, and promotion status.
 7. Create a **5-parameter constructor** with all employee details.
 8. Initialize all employee details using the constructor parameters.
 9. Create a display() method to print employee information.
 10. In main(), create object e1 using the default constructor.
 11. Create object e2 using the 2-parameter constructor.
 12. Create object e3 using the 5-parameter constructor.
 13. Call display() for all three objects.
 14. Display the employee details.
 15. End the program.
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2. Overall Algorithm

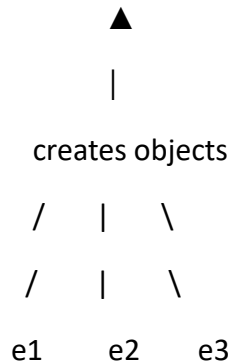
1. **Start**

2. Define the Employee class.
3. Declare employee attributes: name, ID, designation, salary, and promotion status.
4. Define a default constructor and assign default values.
5. Define a constructor with name and id.
6. Define a constructor with all five employee parameters.
7. Define the display() method.
8. Create e1 using the default constructor.
9. Create e2 using the constructor with two parameters.
10. Create e3 using the constructor with five parameters.
11. Call display() for e1.
12. Call display() for e2.
13. Call display() for e3.
14. Display all employee details.
15. **Stop.**

3. UML Diagram :



```
| promotionStatus : String)      |
| + display() : void            |
|                               |
```



4.Code :

```
class Employee {
```

```
    String name;
```

```
    int id;
```

```
    String designation;
```

```
    double salary;
```

```
    String promotionStatus;
```

```
// Default constructor
```

```
Employee() {
```

```
    name = "Unknown";
```

```
    id = 0;
```

```
    designation = "Not Assigned";
```

```
    salary = 0;
```

```
    promotionStatus = "Pending";
```

```
}
```

```
// Constructor with 2 parameters
```

```
Employee(String name, int id) {
```

```
this.name = name;

this.id = id;

designation = "Employee";

salary = 20000;

promotionStatus = "Not Eligible";
}
```

```
// Constructor with 5 parameters
```

```
Employee(String name, int id, String designation,
        double salary, String promotionStatus) {
```

```
    this.name = name;

    this.id = id;

    this.designation = designation;

    this.salary = salary;

    this.promotionStatus = promotionStatus;
}
```

```
// Method to display employee details
```

```
void display() {

    System.out.println("Name      : " + name);

    System.out.println("ID        : " + id);

    System.out.println("Designation  : " + designation);

    System.out.println("Salary      : " + salary);

    System.out.println("Promotion Status : " + promotionStatus);

    System.out.println();

}
}
```

```
public class EmployeeDemo {  
    public static void main(String[] args) {  
  
        // Using default constructor  
        Employee e1 = new Employee();  
  
        // Using constructor with 2 parameters  
        Employee e2 = new Employee("Rahul", 101);  
  
        // Using constructor with 5 parameters  
        Employee e3 = new Employee(  
            "Anil", 102, "Manager", 60000, "Promoted"  
        );  
  
        e1.display();  
        e2.display();  
        e3.display();  
    }  
}
```

5.Output :

```
'-cp' 'C:\Users\yabhi\AppData\Roaming\Code\User\w  
'EmployeeDemo'
```

```
Name          : Unknown  
ID            : 0  
Designation    : Not Assigned  
Salary        : 0.0  
Promotion Status : Pending
```

```
Name          : Rahul  
ID            : 101  
Designation    : Employee  
Salary        : 20000.0  
Promotion Status : Not Eligible
```

```
Name          : Anil  
ID            : 102  
Designation    : Manager  
Salary        : 60000.0  
Promotion Status : Promoted
```

```
PS C:\Users\yabhi\abhi.java> █
```